

1. Loan request input

amount, term, defaults, income, transaction activity



2. Backend and ML service

Express.js /api/risk/score to FastAPI /v1/score and /v1/brief

trained model files loaded, or fallback for local testing



3. Scoring pipeline

22-dim CWB feature vector

Random Forest + Isolation Forest + logistic regression stack



4. Risk outputs and explainability

risk score, anomaly score, APPROVE / REVIEW / REJECT

SHAP top-six reasons for approver review



5. Approver brief and on-chain commit

Authority Brief for bank approvers

commit-reveal oracle stores hash on-chain